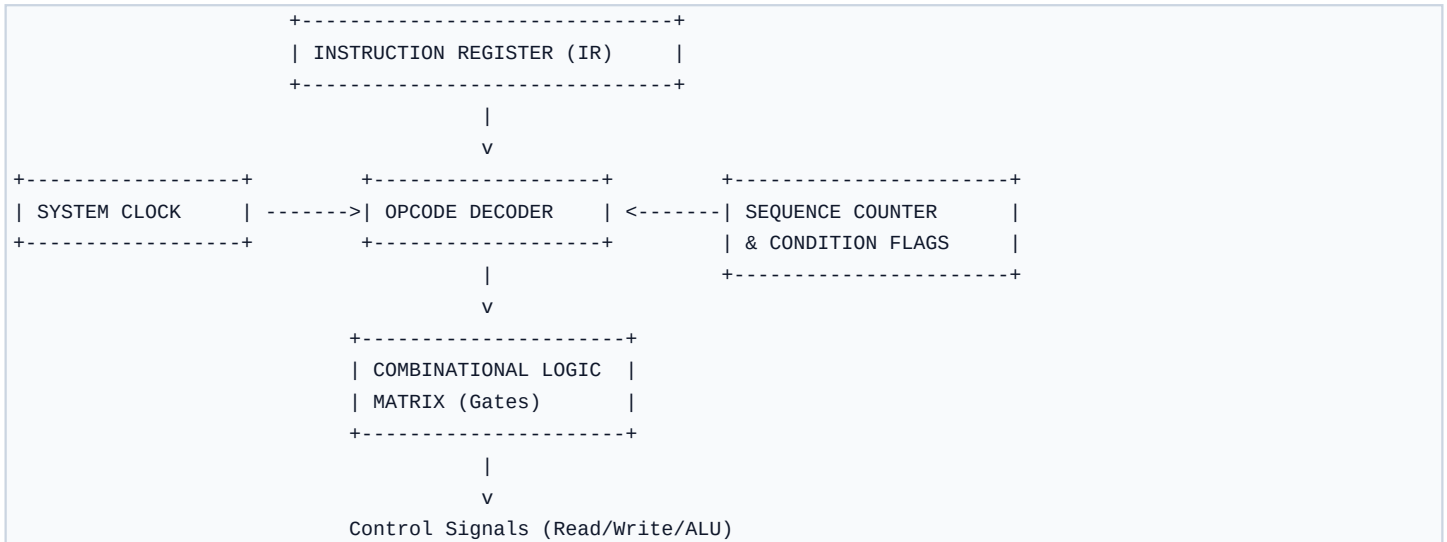


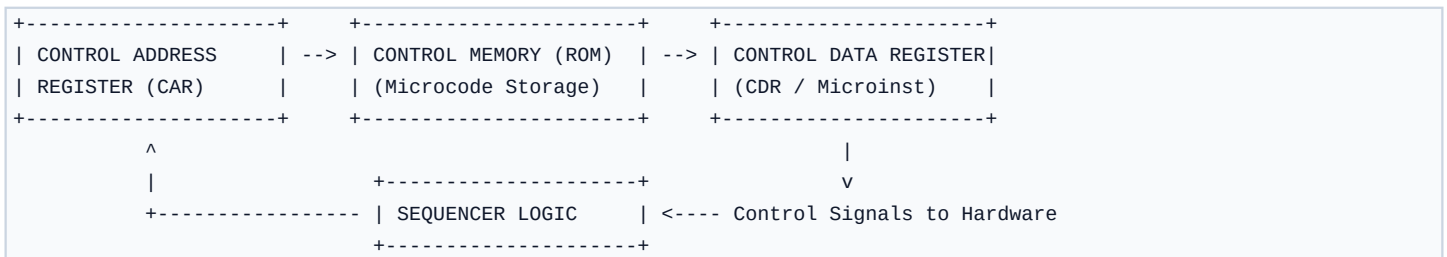
10-Mark Questions & Answers -- MPCA Unit 2: Processor Organization & Architecture

Q1. Explain hardwired and microprogrammed control units in detail with block diagrams, and compare their advantages/disadvantages.

1. Hardwired Control Unit Architecture



2. Microprogrammed Control Unit Architecture



3. Comprehensive Comparison Table

Feature	Hardwired Control Unit	Microprogrammed Control Unit
Logic Implementation	Fixed combinational logic gates & flip	Microcode software stored in Control M
Execution Speed	Extremely fast (limited only by hardwa	Slower (dependent on Control ROM fetch
Flexibility	Rigid. Modifications require hardware	Highly flexible. Upgraded by updating
Instruction Set Type	Best suited for RISC architectures.	Best suited for complex CISC architect
Cost & Complexity	High cost and design complexity for Ia	Systematic design, lower cost for comp
Debugging & Maintenance	Difficult to debug and maintain.	Easier to debug and maintain.

Q2. Explain instruction pipelining in detail. Discuss structural, data, and control hazards with examples and methods to resolve each (forwarding, stalling, branch prediction).

1. 5-Stage RISC Instruction Pipeline

Stage 1: IF (Instruction Fetch)	-> Read instruction from I-Cache.
Stage 2: ID (Instruction Decode)	-> Read registers & decode instruction.
Stage 3: EX (Execution / ALU)	-> Execute ALU operation or calculate memory address.
Stage 4: MEM (Memory Access)	-> Read/Write data from D-Cache.
Stage 5: WB (Write Back)	-> Write result back into destination register.

2. Pipeline Hazards & Resolution Strategies

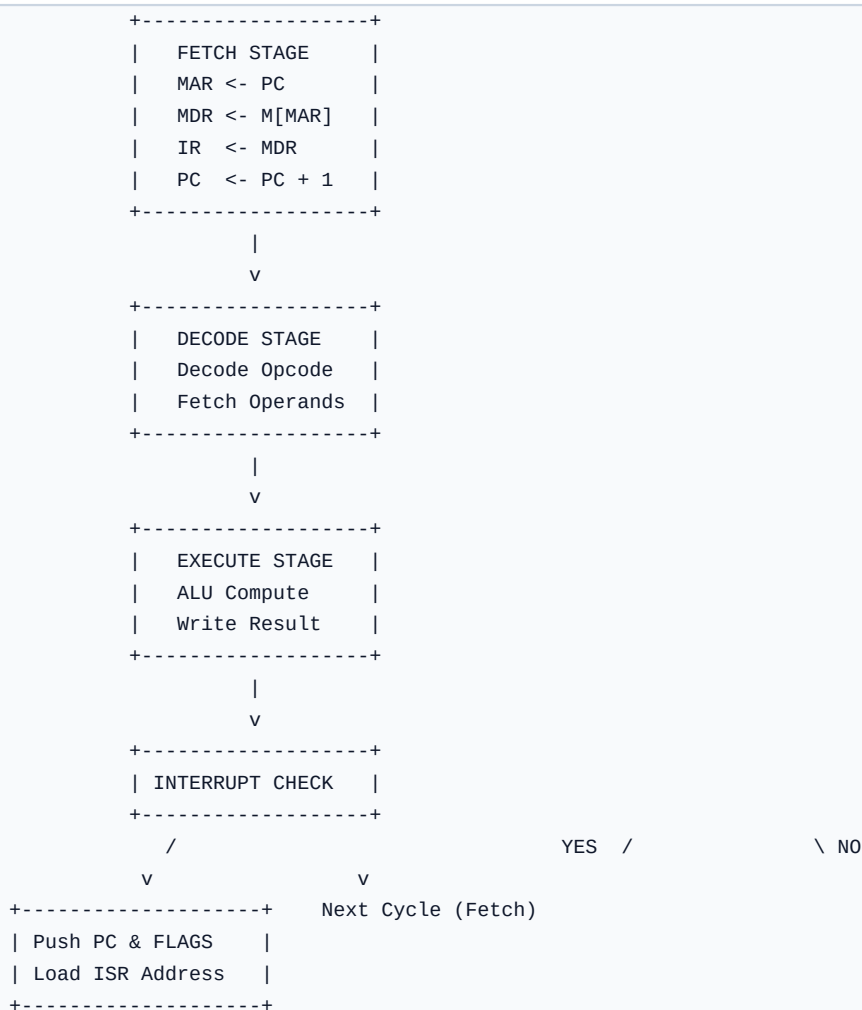
1. Structural Hazards (Resource Contention):

2. Data Hazards (Data Dependencies):

* *Resolution Methods:*

3. Control Hazards (Branch Dependencies):

Q3. Explain the basic instruction cycle with interrupt processing in detail, including the fetch-decode-execute cycle and interrupt handling mechanism.



Q4. Describe microinstruction sequencing and execution techniques (in microprogrammed control) along with a suitable example of microinstruction format.

1. Microinstruction Format Structure

+-----+	+-----+	+-----+	+-----+
CONTROL FIELD	CONDITION FIELD	BRANCH FIELD	ADDRESS FIELD
(Control Signals)	(Status Flags Test)	(Jump Condition)	(Next Micro-Addr)
(16-32 bits)	(2-4 bits)	(2-3 bits)	(8-12 bits)
+-----+	+-----+	+-----+	+-----+

Q5. Explain Flynn's classification of computer architectures in detail with block diagrams and real-world examples for each category.

